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# A long short-term memory enhanced realized conditional heteroskedasticity model

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A B S T R A C T

This paper examines the potential of using realized volatility measures for capturing financial markets' uncertainty. Earlier studies show the usefulness of the high-frequency data based Generalized AutoRegressive Conditional Heteroskedasticity (RealGARCH) model for enhancing volatility forecasting accuracy; however, this model focuses only on linear and short-term dependencies of realized volatility measures on the underlying volatility. Recognizing the critical economic implications of this limitation, the long short-term memory neural network is integrated into RealGARCH, aiming to explore the full impact of realized volatility on volatility modeling and forecasting via capturing the nonlinear and long-term effects. A comprehensive empirical study using 31 indices from 2004 to 2021 is conducted. The results demonstrate that our proposed framework achieves superior in-sample and out-of-sample performance compared to several benchmark models. Importantly, it retains interpretability and effectively adapts to the stylized facts observed in volatility, emphasizing its significant potential for enhancing economic decision-making and risk management.

## 1. Introduction

Volatility modeling is an established area of research in financial econometrics with significant implications for risk management, portfolio allocation, and option pricing; see, e.g., Lehar et al. (2002), Brooks and Persand (2003), Bucci et al. (2023), Liu and Garrett (2023). The Generalized AutoRegressive Conditional Heteroskedasticity (GARCH) model (Engle, 1982; Bollerslev, 1986), and its variants including the exponential GARCH model (Nelson, 1991) and the GJR model (Glosten et al., 1993), provide an excellent modeling framework for understanding and forecasting volatility.

This deficiency can be mitigated by incorporating more effective volatility proxies based on high-frequency intraday return data. In the past two decades, many ex-post estimators of asset return volatility using high-frequency data were introduced in the literature. Examples include the realized variance of Andersen and Bollerslev (1998), realized kernel variance of Barndorff-Nielsen et al. (2008) and bipower variation of Barndorff-Nielsen and Shephard (2004). These estimators, collectively referred to as realized volatility measures, are more informative than daily squared returns about the underlying volatility level, thus providing a better tool for modeling and forecasting volatility. Out article adopts the 5-minute realized variance (RV) of Andersen et al. (2003) as the realized volatility measure. Fig. 1 displays a plot of the daily absolute returns and $\sqrt{RV}$ for the S&P 500. Graphically, absolute returns are comparatively very noisy, whilst $\sqrt{RV}$ are far less noisy proxies.

In a significant development in the GARCH literature, Hansen et al. (2012) developed a joint model, called the realized GARCH (RealGARCH) model, that incorporates a realized measure into a GARCH framework and at the same time includes a measurement equation to model the contemporaneous dependence between the volatility and realized volatility measure dynamics. This model has proven to have superior forecast performance empirically (Jiang et al., 2018; Xie and Yu, 2020; Li et al., 2021; Naimoli et al., 2022). The RealGARCH model expresses the underlying volatility as a *linear* combination of lagged squared returns and realized measures. However, this approach might fall short in capturing the nonlinear relationships and enduring impacts that prior realized measures have on the present intrinsic volatility. This issue is similar to the well-known problem of GARCH that has led to the development of long-memory and nonlinear GARCH models such as Fractionally Integrated GARCH of Baillie et al. (1996) and Recurrent Conditional Heteroskedasticity (RECH) of Nguyen et al. (2022).

This paper aims at examining the *long-range* and *nonlinear* effects that RV has on the underlying volatility dynamics. To this end, the paper develops a flexible volatility model that leverages modern neural network (NN) techniques and incorporates a realized volatility measure. This new model extends RealGARCH and addresses its limitation.

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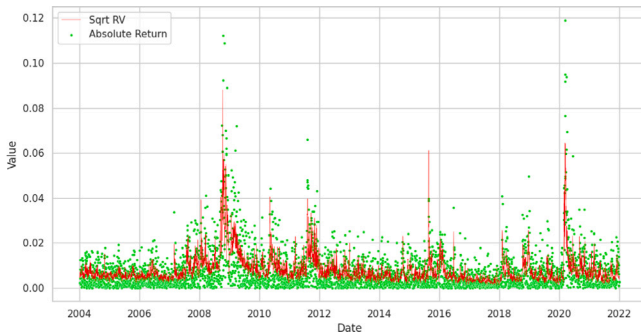


**Fig. 1.** S&P 500 absolute return and $\sqrt{RV}$.

NN is a machine learning technique that is capable of learning complex nonlinear functions and capturing long-range dependence in time series data. With the emergence of NN models and increased computational capabilities, NNs have recently been used in various financial time series forecasting applications. These applications include asset pricing (Fallahgoul et al., 2024), option price forecasting (Almeida et al., 2023), portfolio optimization (Zhang et al., 2020), intraday electricity price forecasting (Klein et al., 2023), as well as volatility and realized measures forecasting (Zhang et al., 2023). Liu (2019) and Bucci (2020) compared the predictive performance of feed-forward neural networks (FNN) and recurrent neural networks (RNN) with traditional econometric approaches, and found that NN models generally outperform econometric models on several stock markets. Yu et al. (2023-12-15) innovatively integrated NN with reinforcement learning, and merged predictions from multiple neural networks, resulting in improved forecasting accuracy. See also Hyup Roh (2007) and Kim and Won (2018). Although these models perform well on specific stock datasets, they are often engineering-oriented and lack interpretability in a financially or economically meaningful way, ignoring important stylized facts such as the leverage and clustering effects commonly observed in financial time series data. An exception is the RECH model of Nguyen et al. (2022), that provides a flexible framework for combining recurrent neural network (RNN) with GARCH-type models. The RECH model represents the volatility as a sum of two components. The first component is governed by a GARCH-type model that retains the characteristics and interpretability of econometric models. The second component is governed by an RNN that can capture nonlinear and long-term serial dependence structure in financial time series. As demonstrated in Nguyen et al. (2022), RECH retains much of the interpretable characteristics from the GARCH model, while enjoying the modeling flexibility and prediction accuracy from RNN. As a development of the basic RNN, the long short-term memory model (LSTM) architecture of Hochreiter and Schmidhuber (1997-11) is one of the most advanced and sophisticated RNN techniques and has proven highly efficient for time series modeling.

There has been little exploration in the literature on how to effectively incorporate realized volatility measures into a NN framework to model volatility, while maintaining the interpretability of the model. To bridge this gap, we introduce a novel LSTM-enhanced realized GARCH framework, under an acronym LSTM-RGARCH, that draws on recent advances in financial econometrics, high-frequency data, and machine learning. We incorporate the LSTM into RealGARCH, extending the usage of simple RNN in RECH. By incorporating LSTM into RealGARCH, we unlock its modeling power and allow it to be able to capture complex underlying dynamics in financial volatility, aiming to model the dynamic nature of volatility more effectively and accurately. Statistical inference and expanding-window sequential prediction in LSTM-RGARCH are performed using recent advancements in Bayesian computation.

To examine the full effect of RV on the underlying volatility, we apply the LSTM-RGARCH model for volatility modeling and forecasting to 31 stock market indices. The main findings can be summarized as follows. First, our analysis provides robust evidence, demonstrated through in-sample fit metrics, that the integration of RV and LSTM techniques substantially enhances the accuracy of the LSTM-RGARCH model in capturing the dynamics of market volatility. The marginal likelihood estimates indicate that, compared to alternative models, the LSTM-RGARCH model offers a superior fit for 28 out of the 31 indices. Through analyzing the posterior distribution of the estimated parameters, we find the presence of nonlinear effects of the lagged realized measures on future volatility. Second, we find strong evidence via out-of-sample testing that incorporating RV and LSTM improves volatility forecasting for various predictive scores. This improvement in forecasting capabilities underscores the potential of combining traditional econometric techniques with modern machine learning to refine economic forecasts and strategic decision-making in financial markets.

The rest of the article is organized as follows. Section 2 briefly reviews the model components (the GARCH-type models, realized volatility measures and recurrent neural networks) before describing the proposed LSTM-RGARCH model. Section 3 presents the Bayesian inference method for LSTM-RGARCH. Section 4 presents the empirical analysis and its econometrics interpretations. Section 5 concludes. The Appendix provides further implementation and empirical details, including a simulation study on option trading. The code and examples reported in the paper are at https://github.com/VBayesLab/DeepRGARCH.

## 2. Realized volatility and model formulation

### 2.1. Conditional heteroscedastic models and realized volatility measures

Let $\boldsymbol{y} = \{y_t, t = 1, \dots, T\}$ be a time series of daily returns. The key term of interest in volatility modeling are the conditional variances, $\sigma_t^2 = \mathrm{var}\left(y_t \mid \mathcal{F}_{t-1}\right)$, where $\mathcal{F}_{t-1}$ denotes the $\sigma$-field of information up to and including time $t-1$. We assume here that $\mathbb{E}(y_t|\mathcal{F}_{t-1}) = 0$, but the present method is easily extended to model the conditional mean $\mathbb{E}(y_t|\mathcal{F}_{t-1})$. The GARCH model expresses the conditional variance $\sigma_t^2$ as a linear combination of the previous squared returns and conditional variances as an ARMA($p$, $q$) model:

$$y_t = \sigma_t \epsilon_t, \quad \epsilon_t \overset{\text{i.i.d}}{\sim} \mathcal{N}(0,1), \quad t = 1, 2, \dots, T \tag{1}$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i y_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2, \quad t = p+1, \dots, T. \tag{2}$$

The restriction $\omega > 0, \alpha_i, \beta_j \geq 0, i = 1, \dots, p, j = 1, \dots, q$ is used to ensure the positivity of $\sigma_t^2$, and $\sum_{i=1}^{p} \alpha_i + \sum_{j=1}^{q} \beta_j < 1$ is needed to ensure the stationarity of the time series $y_t$ (Bollerslev, 1986). The errors $\epsilon_t$ are independently and identically distributed as normal distributions with zero mean and unit variance; other distributions for $\epsilon_t$ such as Student's $t$ are also considered in the literature, e.g., Gerlach and Wang (2016). For other GARCH-type models, the reader is referred to Nelson (1991), Glosten et al. (1993) and Bollerslev (2008).

The GARCH omits intraday information while it is widely known in the financial econometric literature that high-frequency return data, such as 5-minute data, can be used to estimate daily volatility with high accuracy. In the past twenty years, many ex-post estimators of daily volatility using high-frequency data were developed and are referred to as "realized measures" (Andersen and Bollerslev, 1998; Barndorff-Nielsen and Shephard, 2004; Barndorff-Nielsen et al., 2008). Engle (2002) was among the first to integrate these ex-post realized volatility measures introduced by Andersen and Bollerslev (1998) into the GARCH model for predicting future volatility. Since then, many volatility models incorporating realized measures have been developed, e.g., Forsberg and Bollerslev (2002), Engle and Gallo (2006), Corsi (2009), Shephard and Sheppard (2010). The RealGARCH of Hansen et al. (2012)

$$y_t = \sigma_t \epsilon_t, \quad t = 1, 2, \dots, T \tag{3a}$$

$$\sigma_t^2 = \omega + \gamma \mathrm{rv}_{t-1} + \beta \sigma_{t-1}^2 \tag{3b}$$

$$\mathrm{rv}_t = \xi + \varphi \sigma_t^2 + \tau\left(\epsilon_t\right) + u_t \tag{3c}$$

is an important development in this direction. Here $\epsilon_t \overset{i.i.d}{\sim} N(0,1)$, $u_t \overset{i.i.d}{\sim} N\left(0, \sigma_u^2\right)$, $\mathrm{rv}_t$ is a realized volatility measure, and $\tau(\epsilon)$ is regarded as the leverage function and used to capture the leverage effect often observed in volatility. Hansen et al. (2012) set $\tau(\epsilon) = \tau_1 \epsilon + \tau_2\left(\epsilon^2 - 1\right)$. An attractive feature of the RealGARCH model is that it contains the measurement Eq. (3c) that accounts for the variation in the realized volatility measure $\mathrm{rv}_t$. It associates the observed realized volatility measure with the underlying latent volatility, in which the integrated high-frequency variance $\mathrm{rv}_t$ is explained as a linear combination of $\sigma_t^2$ plus a random innovation.

Our article employs a simple yet effective realized volatility measure, the 5-minute realized variance ($\mathrm{RV}_5$) of Andersen and Bollerslev (1998). Suppose that we observe the asset price at $n$ trading times within a trading day $t$, $t_j = t - 1 + j/n, j = 1, \dots, n$. Let $\{P(t_j), j = 1, \dots, n\}$ be the observed prices and $r_{t_j} = \log P\left(t_j\right) - \log P\left(t_{j-1}\right)$ be the log-returns. The RV for the trading day $t$ is defined as

$$\mathrm{rv}_t := \sum_{j=1}^{n} r_{t_j}^2 .$$

It can be shown that Andersen and Bollerslev (1998), as $n \to \infty$, $\mathrm{rv}_t$ converges in probability to the true latent variance $\sigma_t^2$. For $\mathrm{RV}_5$, the return $r_{t_j}$ in the above equation is recorded at 5 min frequency. There are various definitions of realized volatility measures but there is little evidence that any outperform $\mathrm{RV}_5$ as a volatility proxy; see Liu et al. (2015) for a detailed comparison of more than 400 realized volatility measures.

### 2.2. Recurrent neural network

RNN is a special class of neural network designed for modeling sequential data. Let $\{D_t = (x_t, y_t), t = 1, 2, \dots\}$ be the data with $x_t$ the input and $y_t$ the output. We use $(x_t, y_t)$ in this section as generic notation, not necessarily applicable to the return data in other sections. The task is to model the conditional mean $\hat{y}_t = \mathbb{E}(y_t | x_t, D_{1:t-1})$. The basic RNN framework is

$$h_t = g_h\left(W^h\left[h_{t-1}, x_t\right] + b^h\right), \tag{4a}$$

$$\hat{y}_t = g_y\left(W^y h_t + b^y\right). \tag{4b}$$

The main feature of the RNN structure is its vector of hidden states $h_t$ which is defined recurrently. At each time $t$, two information sources are fed into $h_t$: the historical information stored in $h_{t-1}$ and the current information from the input $x_t$. The functions $g_h$ and $g_y$ are activation functions such as $\mathrm{sig}(z) := 1/(1+e^{-z})$, or $\tanh(z) := (e^z - e^{-z})/(e^z + e^{-z})$. Finally, the $W$ and $b$ are trainable model parameters.

The basic RNN model in (4a)–(4b) has some limitations in terms of both modeling flexibility and training difficulty. Many sophisticated RNN structures are proposed to overcome these limitations, and the LSTM model of Hochreiter and Schmidhuber (1997-11) stands out as one of the most successful methods. LSTM uses a gate structure to control the memory in the data. It is written as follows:

$$g_t^i = \mathrm{sig}\left(W^i\left[h_{t-1}, x_t\right] + b^i\right) \tag{5a}$$

$$g_t^f = \mathrm{sig}\left(W^f\left[h_{t-1}, x_t\right] + b^f\right) \tag{5b}$$

$$g_t^o = \mathrm{sig}\left(W^o\left[h_{t-1}, x_t\right] + b^o\right) \tag{5c}$$

$$\tilde{c}_t = \tanh\left(W^c\left[h_{t-1}, x_t\right] + b^c\right) \tag{5d}$$

$$c_t = g_t^i \cdot \tilde{c}_t + g_t^f \cdot c_{t-1} \tag{5e}$$

$$h_t = g_t^o \cdot \tanh\left(c_t\right) \tag{5f}$$

$$\hat{y}_t = g_y\left(W^y h_t + b^y\right). \tag{5g}$$

Unlike the basic RNN that fully overwrites the memory stored in the hidden states at each step, LSTM can decide to keep, forget or update the memory via the memory cell $c_t$ in (5e). This memory cell $c_t$ is updated by partially forgetting the previous memory from $c_{t-1}$ and adding new memory from $\tilde{c}_t$. The extent of forgetting the history and adding new information is controlled by the forget gate $g_t^f$ and input gate $g_t^i$, respectively. Finally, the degree of current memory usage for the final output is controlled by the output gate $g_t^o$. See Goodfellow et al. (2016), for a book length discussion of RNN and deep learning methods in general.

The ability of LSTM to control its memory and quickly adapt to new data patterns makes it suitable to model volatility dynamics. It is well-known that volatility dynamics has long memory as well as exhibiting clustering (Mandelbrot, 1967; Ding et al., 1993; Christensen and Nielsen, 2007). Section 2.3 describes how to incorporate LSTM into volatility modeling.

### 2.3. LSTM enhanced realized GARCH

This section presents our LSTM-RGARCH model, that incorporates LSTM into RealGARCH for flexible and accurate volatility modeling. Its key motivation is using an additive component governed by LSTM, that takes the realized measure as its input, to capture any complex serial dependence structure in the volatility dynamics. As intraday realized volatility measures are accurate volatility proxies (Andersen and Bollerslev, 1998; Barndorff-Nielsen et al., 2008), feeding them to the LSTM unit further strengthens its modeling capacity. The LSTM-RGARCH model of order $(p, q) = (1, 1)$, LSTM-RGARCH(1,1), is written as:

$$y_t = \sigma_t \epsilon_t, \quad t = 1, 2, \dots \tag{6a}$$

$$\sigma_t^2 = g(\omega_t) + \gamma \mathrm{rv}_{t-1} + \beta \sigma_{t-1}^2 \tag{6b}$$

$$\mathrm{rv}_t = \xi + \varphi \sigma_t^2 + \tau\left(\epsilon_t\right) + u_t \tag{6c}$$

$$\omega_t = \beta_0 + \beta_1 h_t \tag{6d}$$

$$h_t = \mathrm{LSTM}\left(x_t\right) \tag{6e}$$

$$x_t = \left(\omega_{t-1}, y_{t-1}, \sigma_{t-1}^2, \mathrm{rv}_{t-1}\right). \tag{6f}$$

Compared to the RealGARCH model, Eq. (3b), that only allows a linear and short-term dependence of the true latent conditional variance $\sigma_t^2$ on the realized volatility measure, the LSTM-RGARCH model is much more flexible. The latter, via its LSTM component $g(\omega_t)$, allows for both nonlinear and long-term dependence that the previous realized volatility measures might have on $\sigma_t^2$. As $\sigma_{t-1}^2$ is also an input to the LSTM component, LSTM-RGARCH also allows for complex serial dependence that the previous volatility might have on $\sigma_t^2$.

This integration of LSTM and econometric models is driven by the observation that NN models offer more explanatory power than econometric models but demand a significant amount of data for training. However, given the restricted size of financial time series, pure NN models often suffer from overfitting. To address this issue, we include the LSTM as an additive "adjustment" to the reliable linear effects of established econometric models. This approach aims to strike a balance between the strengths of NN and mitigate the risk of overfitting.

In addition, there is a debate on how NN can be employed in financial time series and volatility modeling (Makridakis et al., 2018), as the NN component may reduce the model interpretability, compared to conventional financial econometrics models. We address this issue by carefully tailoring the LSTM structure to enhance the realized GARCH framework. Several key parts of the original RealGARCH model, e.g., the leverage component and the measurement equation are kept, to maintain interpretability. As Table 2 in Section 4.1, the parameter estimates of the proposed model can be directly compared with ones from the conventional volatility models, so that we can interpret how the new framework behaves compared to the existing models and understand how the proposed framework can improve volatility forecasting.

$g$ in (6b) is a non-negative activation function, applied to the $\omega_t$ to ensure a positive conditional variance. We adopt the RELU function, $g(\omega_t) := \max\{\omega_t, 0\}$, for this purpose. Inspired by RealGARCH, LSTM-RGARCH includes the measurement Eq. (6c) to account for the variation in the realized volatility measures. To prevent overfitting, we have used a LSTM model featuring a single layer and one hidden state. Rather than using the linear structure as in (6c), one can easily use an RNN model to explain $\mathrm{rv}_t$ based on $\sigma_t^2$; we tried this, but did not observe any meaningful improvement.

Following Amazon's DeepAR model (Salinas et al., 2020), beside $y_{t-1}$ and $rv_{t-1}$, we also incorporated model intermediates and outputs, namely $\omega_{t-1}$ and $\sigma_{t-1}$, into the input vector $x_t$. This adaptation effectively transforms the NN component of LSTM-RGARCH in an autoregressive fashion. Although our paper uses only four input features, it is straightforward to expand this by including different realized volatility measures or other features such as open ($o_{t-1}$), high ($hi_{t-1}$), low ($l_{t-1}$), and volume ($v_{t-1}$), as supplementary inputs:

$$x_t = \left(\omega_{t-1}, y_{t-1}, \sigma_{t-1}^2, o_{t-1}, hi_{t-1}, l_{t-1}, v_{t-1}, \mathrm{rv}_{t-1}^1, \mathrm{rv}_{t-1}^2 \dots\right), \tag{7}$$

$\mathrm{rv}^1$ and $\mathrm{rv}^2$ denote different types of realized measures. For an in-depth examination of influential features in forecasting equity series, readers are encouraged to explore the extensive analysis conducted by Feng et al. (2020). Although our paper focuses on the LSTM-RGARCH(1,1) model, it is noteworthy that the model's flexibility extends to LSTM-RGARCH($p, q$) by replacing Eq. (6b) with:

$$\sigma_t^2 = g(\omega_t) + \sum_{i=1}^{p} \gamma_i \mathrm{rv}_{t-i} + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2. \tag{8}$$

## 3. Bayesian inference for the LSTM-RGARCH model

### 3.1. The likelihood and prior

We adopt a Bayesian inference approach to estimate the LSTM-RGARCH model. Following Hansen et al. (2012), we assume Gaussian errors $\epsilon_t \overset{i.i.d}{\sim} N(0,1)$ and $u_t \overset{i.i.d}{\sim} N(0, \sigma_u)$. Hence, given the observed data ($\boldsymbol{y} = \{y_1, \dots, y_T\}, \boldsymbol{rv} = (\mathrm{rv}_1, \dots, \mathrm{rv}_T)$) the log-likelihood function of the LSTM-RGARCH model is

$$\ell(y, rv|\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log\left(\sigma_t^2\right) + y_t^2/\sigma_t^2\right] - \frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log\left(\sigma_u^2\right) + u_t^2/\sigma_u^2\right], \tag{9}$$

where $u_t = \mathrm{rv}_t - \xi - \varphi\sigma_t^2 - \tau_1 y_t/\sigma_t - \tau_2\left(y_t^2/\sigma_t^2 - 1\right)$. Recall that the vector of model parameters $\theta$ consists of the GARCH and LSTM parameters. For example, the LSTM-RGARCH(1,1) model with a single hidden state LSTM has 7 GARCH parameters, $(\beta, \gamma, \xi, \varphi, \tau_1, \tau_2, \sigma_u)$, and 26 LSTM parameters including $\beta_0$, $\beta_1$ and 24 parameters within the LSTM structure.

We apply commonly used priors to the GARCH parameters in the literature; see, e.g., Gerlach and Wang (2016). The prior $N(0, 0.1)$ is used for all the LSTM parameters.

### 3.2. Model estimation and prediction

For Bayesian inference and prediction in volatility modeling, the Sequential Monte Carlo (SMC) method (Neal, 2001; Chopin, 2002; Del Moral et al., 2012) is an effective approach for computing expanding-window volatility forecasts which can effectively sample from non-standard posteriors, while also providing the marginal likelihood estimate as a by-product. SMC uses a set of $M$ weighted particles initially sampled from an easy-to-sample distribution, such as the prior $p(\theta)$, with the particles then traversed through intermediate distributions which eventually become the target distribution. See Gunawan et al. (2022-10-10) for a review of the SMC method.

For in-sample model estimation and inference, we use the likelihood annealing version of SMC that samples from the sequence of distributions

$$\pi_k(\theta) \propto p(\theta)p(\boldsymbol{y}, \boldsymbol{rv} \mid \theta)^{\gamma_k}, \quad k = 0, 1, \dots, K; \tag{10}$$

here, $0 = \gamma_0 < \gamma_1 < \gamma_2 < \cdots < \gamma_K = 1$ are called the temperature levels. Reweighting, resampling, and a Markov transition are the three primary components of the SMC approach. Several methods exist to implement SMC in practice, and we briefly describe one of them now. The collection of weighted particles $\left\{W_{k-1}^j, \theta_{k-1}^j\right\}_{j=1}^M$ that approximate the intermediate distribution $\pi_{k-1}(\theta)$ is reweighted at the start of iteration $k$ to approximate the target $\pi_k(\theta)$. The efficiency of these weighted particles is measured by the effective sample size (ESS) (Kass et al., 1998)

$$\mathrm{ESS} = \frac{1}{\sum_{j=1}^{M}\left(W_t^j\right)^2}, \tag{11}$$

where $W_t := (W_t^1, \dots, W_t^M)$ is the weight vector for the $M$ particles at time $t$. If the ESS is below a threshold, the particles are resampled to obtain equally weighted particles, and a Markov kernel with the invariant distribution $\pi_k(\theta)$ is then applied to refresh these equally weighted particles. Following Del Moral et al. (2012), we adaptively choose the tempering sequence $\gamma_k$ in order to maintain an adequate particle diversity.

For out-of-sample expanding-window volatility forecasting at each time $T + t$, $t = 1, 2, \dots$, it is necessary to infer the posterior predictive distribution of $(y_{T+t+1}, \boldsymbol{rv}_{1:T+t+1})$ based on the data $(y_{1:T+t}, \boldsymbol{rv}_{1:T+t})$. This can be performed by employing the data annealing SMC sampler that samples from the sequence

$$\pi_t(\theta) = p(\theta|y_{1:T+t}, \boldsymbol{rv}_{1:T+t}) \propto p(\theta)p(\boldsymbol{y}_{1:T+t}, \boldsymbol{rv}_{1:T+t}|\theta), \quad t = 1, 2, \dots \tag{12}$$

## 4. Empirical analysis

This section studies the performance of the LSTM-RGARCH(1,1) model and compares it to GARCH, RealGARCH and RECH using 31 stock indices including the Amsterdam Exchange Index (AEX), the Dow Jones Index (DJI), the Frankfurt Stock Exchange (GDAXI) and the Standard and Poor's 500 Index (SP500). The datasets were downloaded from the Realized Library of The Oxford-Man Institute. In the main text, we present results for the above mentioned four representative indices and the average over 31 indices to conserve space; the detailed results for all 31 indices are shown in Appendix 6.2. Given the in-sample daily closing prices $\{P_t, t = 0, \dots, T_{in}\}$, we compute the demeaned close-to-close return process as

$$y_t = 100\left(\log\frac{P_t}{P_{t-1}} - \frac{1}{T_{in}}\sum_{t=1}^{T_{in}}\log\frac{P_t}{P_{t-1}}\right), \quad t = 1, 2, \dots, T_{in} \tag{13}$$

where $T_in$ is the number of in-sample observations. We adopt the 5 mins realized variance (Andersen et al., 2003) as the realized volatility measure, $\mathrm{rv}_t$, in RealGARCH and LSTM-RGARCH. As the realized volatility measures ignore the overnight variation of the prices and sometimes the variation in the first few minutes of the trading day when recorded prices may contain large errors (Shephard and Sheppard, 2010), we follow Hansen and Lunde (2005) and scale the realized volatility measure as

$$\tilde{\sigma}_t^2 = \hat{c} \cdot \mathrm{rv}_t \text{ where } \hat{c} = \frac{\sum_{t=1}^{T_{in}} y_t^2}{\sum_{t=1}^{T_{in}} \mathrm{rv}_t}, \quad t = 1, 2, \dots, T_{in}. \tag{14}$$

and use $\tilde{\sigma}_t^2$ as the estimate of the latent conditional variance $\sigma_t^2$.

The date range is from January 2004 to December 2021, including the COVID-19 pandemic period, which we divide it into a first half

**Table 1**
Dataset specifications, ranked by Market Cap. The last column presents the estimated market capitalization of each index in USD trillion, as of the information available until the end of 2023.

| | In-Sample | Out-of-Sample | T_in | T_out | Market Cap |
|---|---|---|---|---|---|
| SPX | 2004-01-02 – 2012-12-27 | 2012-12-28 – 2021-12-31 | 2259 | 2259 | 42.0 |
| IXIC | 2004-01-02 – 2012-12-31 | 2013-01-02 – 2021-12-31 | 2260 | 2261 | 20.0 |
| DJI | 2004-01-02 – 2012-12-26 | 2012-12-27 – 2021-12-31 | 2258 | 2258 | 11.0 |
| SSEC | 2004-01-02 – 2012-12-31 | 2013-01-04 – 2021-12-31 | 2184 | 2185 | 7.0 |
| N225 | 2004-01-05 – 2012-12-14 | 2012-12-17 – 2021-12-30 | 2198 | 2199 | 6.0 |
| HSI | 2004-01-02 – 2012-12-14 | 2012-12-17 – 2021-12-31 | 2213 | 2214 | 5.0 |
| STOXX50E | 2004-01-02 – 2012-12-24 | 2012-12-27 – 2021-12-31 | 2292 | 2292 | 3.8 |
| NSEI | 2004-01-01 – 2012-12-21 | 2012-12-24 – 2021-12-31 | 2232 | 2233 | 3.5 |
| FTSE | 2004-01-02 – 2013-01-03 | 2013-01-04 – 2021-12-31 | 2273 | 2273 | 3.0 |
| BSESN | 2004-01-01 – 2012-12-20 | 2012-12-21 – 2021-12-31 | 2233 | 2234 | 3.0 |
| GSPTSE | 2004-01-02 – 2013-01-03 | 2013-01-04 – 2021-12-31 | 2250 | 2250 | 2.3 |
| FCHI | 2004-01-02 – 2012-12-24 | 2012-12-27 – 2021-12-31 | 2303 | 2304 | 2.5 |
| RUT | 2004-01-02 – 2012-12-28 | 2012-12-31 – 2021-12-31 | 2259 | 2259 | 2.4 |
| GDAXI | 2004-01-02 – 2012-12-10 | 2012-12-11 – 2021-12-30 | 2282 | 2282 | 2.0 |
| KS11 | 2004-01-02 – 2012-12-07 | 2012-12-10 – 2021-12-30 | 2223 | 2223 | 1.8 |
| SSMI | 2004-01-05 – 2012-12-13 | 2012-12-14 – 2021-12-30 | 2258 | 2258 | 1.6 |
| AORD | 2004-01-02 – 2013-01-02 | 2013-01-03 – 2021-12-31 | 2276 | 2277 | 1.4 |
| BVSP | 2004-01-02 – 2012-12-17 | 2012-12-18 – 2021-12-30 | 2216 | 2216 | 1.2 |
| IBEX | 2004-01-02 – 2013–0104 | 2013-01-07 – 2021-12-30 | 2292 | 2292 | 0.8 |
| FTMIB | 2009-06-01 – 2015-09-08 | 2015-09-09 – 2021-12-30 | 1594 | 1595 | 0.7 |
| AEX | 2004-01-02 – 2012-12-21 | 2012-12-24 – 2021-12-31 | 2302 | 2302 | 0.6 |
| OMXSPI | 2005-10-03 – 2013-11-08 | 2013-11-11 – 2021-12-30 | 2037 | 2037 | 0.5 |
| MXX | 2004-01-02 – 2012-12-20 | 2012-12-21 – 2021-12-31 | 2261 | 2261 | 0.5 |
| BFX | 2004-01-02 – 2012-12-21 | 2012-12-24 – 2021-12-31 | 2302 | 2302 | 0.5 |
| STI | 2004-01-02 – 2016-11-03 | 2016-11-04 – 2021-12-31 | 1289 | 1289 | 0.4 |
| OSEAX | 2004-01-02 – 2012-12-06 | 2012-12-07 – 2021-12-30 | 2247 | 2247 | 0.3 |
| OMXC20 | 2005-10-03 – 2013-11-14 | 2013-11-15 – 2021-12-30 | 2018 | 2018 | 0.3 |
| OMXHPI | 2005-10-03 – 2013-11-07 | 2013-11-08 – 2021-12-30 | 2037 | 2037 | 0.2 |
| BVLG | 2012-10-15 – 2017-05-18 | 2017-05-21 – 2021-12-31 | 1171 | 1172 | 0.1 |
| SMSI | 2005-07-04 – 2013-09-29 | 2013-09-30 – 2021-12-30 | 2101 | 2102 | 0.1 |
| KSE | 2004-01-01 – 2013-02-11 | 2013-02-12 – 2021-12-31 | 2196 | 2196 | 0.07 |

for training and the second half for out of sample analysis. Table 1 provides a breakdown of each index, including their respective market capitalization. To keep the models current and utilize historical data efficiently, a daily expanding window forecasting approach is implemented.

### *4.1. Parameter estimation and in-sample fit*

As mentioned in Section 3, the marginal likelihood estimate is a by-product of SMC, which is useful for in-sample model comparison using a Bayes factor. Given the training data ($\boldsymbol{y} = y_{1:T}, \boldsymbol{rv} = \mathrm{rv}_{1:T}$), let $\widehat{p}\left(\boldsymbol{y}, \boldsymbol{rv} | M_i\right)$ be the marginal likelihood estimate of model $M_i$, $i = 1, 2$. The Bayes factor of model $M_1$ relative to $M_2$ is

$$\mathrm{BF}_{M_1, M2} = \frac{\widehat{p}\left(\boldsymbol{y}, \boldsymbol{rv} | M_1\right)}{\widehat{p}\left(\boldsymbol{y}, \boldsymbol{rv} | M_2\right)}. \tag{15}$$

The larger the Bayes factor, the more decisive is the evidence that model $M_1$ is superior to $M_2$ (Kass and Raftery, 1995). Table 2 summarizes the parameter estimation (for the five main parameters) and the Bayes factor (with GARCH used as the baseline model) for the four models, GARCH, RealGARCH, RECH and LSTM-RGARCH. The last panel reports the results for the average over the 31 indices.

We draw the following conclusions from the estimation results. First, the marginal likelihood estimates show that the LSTM-RGARCH model fits the index datasets better than the competing models for 28 of 31 indices. On average, the Bayes factors of the LSTM-RGARCH model compared with the GARCH, RECH, and RealGARCH models are roughly $e^{58}$, $e^{30}$ and $e^{14}$, respectively, which according to Jeffery's scale for interpreting the Bayes factor (Jeffreys, 1998), decisively support the LSTM-RGARCH model. Second, the estimated posterior mean of the parameter $\beta_1$ in the LSTM-RGARCH model is significantly different from zero in all cases, providing evidence of the volatility effects rather than linearity that the previous conditional variance $\sigma^2_{t-1}$ and realized volatility measure $\mathrm{rv}_{t-1}$ have on $\sigma^2_t$. This also suggests that the LSTM component $g(\omega_t)$ in LSTM-RGARCH can detect these effects effectively.

Third, as discussed in Section 2.3, the LSTM component is carefully tailored to enhance the realized GARCH framework and build a hybrid model that we expect to offer enhanced interpretability compared to black-box NN models. We now discuss the parameter estimates in detail (using the "Mean" estimates in Table 2), to show how LSTM-RGARCH behaves differently to the existing models and explain why LSTM-RGARCH achieves improved volatility forecasts. Our discussion focuses on two points: the importance of incorporating the LSTM component to capture the nonlinear and long term time series dependence (development of LSTM-RGARCH compared to RealGARCH); the effectiveness of including the high frequency data based realized volatility measure in the LSTM component (development of LSTM-RGARCH comparing to RECH). First, the estimate $\gamma$ (the coefficient for the lagged realized volatility measure) of RealGARCH is larger than the estimate of $\alpha$ (the coefficient for the lagged return) from GARCH (0.306 vs 0.102). This demonstrates the effectiveness of employing the high frequency data based realized measure for volatility forecasting, as such a measure is more informative and efficient than the daily return. Furthermore, comparing GARCH and RECH, both the $\alpha$ and $\beta$ estimates of RECH are on average smaller than that for GARCH. Such a finding shows that the LSTM based time varying intercept term $g(\omega_t)$ contains more information than a constant intercept $\omega$. Last, focusing on the proposed LSTM-RGARCH model, its $\gamma$ estimate (the coefficient for the lagged realized volatility measure) is smaller than that of RealGARCH. This means that more information (again nonlinear and long term dependence) is contained in the LSTM and high frequency data based time varying intercept term $g(\omega_t)$. Comparing to RECH, LSTM-RGARCH employs the lagged realized volatility measure to drive the future volatility dynamics (via $\gamma$) and as the input of the LSTM component (via $g(\omega_t)$). The LSTM-RGARCH $\gamma$ estimate is 0.168 which is larger than the $\alpha$ estimate (0.059) in RECH, supporting the importance of using the realized measure in volatility forecasting.

**Table 2**
In-sample analysis: Posterior means of the parameters with the posterior standard deviations (in parentheses).

| | | $\alpha$ | $\beta$ | $\beta_0$ | $\beta_1$ | $\gamma$ | Mar.llik | BF |
|---|---|---|---|---|---|---|---|---|
| AEX | GARCH | 0.105 (0.012) | 0.884 (0.013) | – | – | – | −3450.8 (0.183) | 1 |
| | RECH | 0.038 (0.014) | 0.741 (0.061) | 0.049 (0.019) | 0.907 (0.182) | – | −3400.6 (0.280) | $e^{50}$ |
| | RealGARCH | – | 0.549 (0.028) | – | – | 0.371 (0.028) | −3376.3 (0.018) | $e^{75}$ |
| | LSTM-RGARCH | – | 0.527 (0.021) | 0.112 (0.022) | 0.801 (0.081) | 0.341 (0.015) | −3370.5 (0.220) | $\mathbf{e^{80}}$ |
| DJI | GARCH | 0.094 (0.011) | 0.890 (0.012) | – | – | – | −3027.2 (0.100) | 1 |
| | RECH | 0.049 (0.011) | 0.636 (0.063) | 0.063 (0.019) | 0.912 (0.182) | – | −2991.0 (0.785) | $e^{36}$ |
| | RealGARCH | – | 0.634 (0.024) | – | – | 0.318 (0.027) | −2968.4 (0.075) | $e^{59}$ |
| | LSTM-RGARCH | – | 0.840 (0.014) | 1.413 (0.148) | 11.827 (0.577) | 0.005 (0.004) | −2962.9 (0.421) | $\mathbf{e^{64}}$ |
| GDAXI | GARCH | 0.098 (0.013) | 0.889 (0.014) | – | – | – | −3615.5 (0.107) | 1 |
| | RECH | 0.049 (0.013) | 0.666 (0.071) | 0.079 (0.036) | 1.000 (0.206) | – | −3570.0 (0.349) | $e^{46}$ |
| | RealGARCH | – | 0.439 (0.025) | – | – | 0.452 (0.031) | −3534.8 (0.055) | $e^{81}$ |
| | LSTM-RGARCH | – | 0.414 (0.029) | 0.126 (0.084) | 5.561 (0.409) | 0.307 (0.022) | −3523.6 (0.141) | $\mathbf{e^{92}}$ |
| SPX | GARCH | 0.091 (0.010) | 0.895 (0.011) | – | – | – | −3177.1 (0.158) | 1 |
| | RECH | 0.049 (0.010) | 0.657 (0.059) | 0.071 (0.020) | 0.871 (0.193) | – | −3150.4 (0.491) | $e^{27}$ |
| | RealGARCH | – | 0.633 (0.022) | – | – | 0.317 (0.025) | −3100.1 (0.078) | $\mathbf{e^{77}}$ |
| | LSTM-RGARCH | – | 0.831 (0.016) | 1.549 (0.160) | 11.432 (0.660) | 0.007 (0.005) | −3104.7 (0.205) | $e^{72}$ |
| Mean | GARCH | 0.102 (0.013) | 0.882 (0.015) | – | – | – | −3411.9 (0.132) | 1 |
| | RECH | 0.059 (0.014) | 0.729 (0.057) | 0.071 (0.032) | 0.850 (0.193) | – | −3384.3 (0.226) | $e^{28}$ |
| | RealGARCH | – | 0.628 (0.024) | – | – | 0.306 (0.025) | −3368.4 (0.214) | $e^{44}$ |
| | LSTM-RGARCH | – | 0.667 (0.034) | 0.613 (0.118) | 5.203 (0.453) | 0.168 (0.021) | −3354.0 (0.472) | $\mathbf{e^{58}}$ |

*Note:* The last two columns show the natural logarithms of the estimated marginal likelihood (Mar.llik) with Monte Carlo standard errors (in parentheses) across 6 different runs of the SMC and the Bayes factors (BF) using GARCH as the baseline.

### 4.2. Volatility forecast error compared to ex-post realized volatility measures

We now test the volatility forecasting performance of the LSTM-RGARCH model. The forecast performance is measured by mean squared error (MSE) and mean absolute error (MAE) computed on test data $D_{test}$ of size $T_{test}$

$$\text{MSE} = T_{test}^{-1} \sum_{D_{test}} (\hat{\sigma}_t - \tilde{\sigma}_t)^2 \tag{16}$$

$$\text{MAE} = T_{test}^{-1} \sum_{D_{test}} |\hat{\sigma}_t - \tilde{\sigma}_t| \tag{17}$$

where $\hat{\sigma}_t$ is the one-step-ahead expanding-window forecast of the latent $\sigma_t$ and $\tilde{\sigma}_t$ is the squared root of an ex-post realized volatility measure after rescaling as in (14). We use five ex-post volatility proxies, the Realized Variance (RV), Bipower Variation (BV), Median Realized Volatility (MedRV), Realized Kernel Variance with the Non-flat Parzen kernel (RK-Parzen) and the Tukey-Hanning kernel (RSV). See Shephard and Sheppard (2010) for details about those realized measures.

Tables 3 and 4 summarize the forecast performance of the four models. The last column in each table shows the number of times a model has the best predictive score across the five volatility proxies. The last panel in each table shows the average scores over the 31 indices. Using the five ex-post realized volatility measures as the ground truth, the results show that the LSTM-RGARCH model performs the best in terms of volatility forecasting.

### 4.3. Fitness to return series and tail risk forecast

An attractive feature of GARCH-type models including LSTM-RGARCH is that they model both the volatility and return processes. This feature is crucial to risk management which is one of the most important applications of volatility modeling. For risk management, the key task is to forecast the Value at Risk (VaR) and Expected Shortfall (ES). An $\alpha$-level VaR is the $\alpha$-level quantile of the return distribution, and the $\alpha$-level ES is the conditional expectation of return values that exceed the corresponding $\alpha$-level VaR. Both VaR and ES are the two key risk measures in financial regulation and recommended by the Basel Accord.

To evaluate the quality of the VaR forecast, we adopt the standard quantile loss function

$$\text{Qloss} := \sum_{y_t \in D_{test}} \left(\alpha - I\left(y_t < Q_t^\alpha\right)\right)\left(y_t - Q_t^\alpha\right), \tag{18}$$

where $Q_t^\alpha$ is the forecast of the $\alpha$-level VaR of $y_t$ (Koenker and Bassett, 1978). We note that, given the volatility $\sigma_t$ and the normality assumption of the random shock $\epsilon_t$ in (6a), it is straightforward to compute $Q_t^\alpha$. The quantile loss function is strictly consistent (Fissler and Ziegel,

**Table 3**
Forecast performance: MSE for different realized volatility measures.

| | | RV5 | BV | MedRV | RK-Parzen | RSV | Count |
|---|---|---|---|---|---|---|---|
| AEX | GARCH | 0.151 | 0.142 | 0.141 | 0.209 | 0.190 | 0.0 |
| | RECH | 0.143 | 0.131 | 0.126 | 0.188 | 0.180 | 0.0 |
| | RealGARCH | 0.121 | 0.109 | 0.102 | 0.173 | 0.158 | 0.0 |
| | LSTM-RGARCH | **0.116** | **0.103** | **0.094** | **0.157** | **0.152** | **5.0** |
| DJI | GARCH | 0.203 | 0.174 | 0.177 | 0.202 | 0.274 | 0.0 |
| | RECH | 0.194 | 0.176 | 0.176 | 0.193 | 0.266 | 0.0 |
| | RealGARCH | 0.121 | 0.124 | 0.127 | 0.119 | 0.196 | 0.0 |
| | LSTM-RGARCH | **0.112** | **0.122** | **0.118** | **0.109** | **0.184** | **5.0** |
| GDAXI | GARCH | 0.180 | 0.174 | 0.182 | 0.203 | 0.221 | 0.0 |
| | RECH | 0.159 | 0.154 | 0.160 | 0.180 | 0.200 | 0.0 |
| | RealGARCH | 0.111 | 0.110 | 0.114 | 0.140 | 0.152 | 0.0 |
| | LSTM-RGARCH | **0.106** | **0.104** | **0.105** | **0.132** | **0.146** | **5.0** |
| SPX | GARCH | 0.183 | 0.166 | 0.180 | 0.182 | 0.243 | 0.0 |
| | RECH | 0.174 | 0.163 | 0.172 | 0.172 | 0.237 | 0.0 |
| | RealGARCH | 0.117 | **0.119** | 0.129 | 0.119 | 0.182 | 1.0 |
| | LSTM-RGARCH | **0.112** | 0.121 | **0.125** | **0.113** | **0.176** | **4.0** |
| Mean | GARCH | 0.196 | 0.174 | 0.184 | 0.243 | 0.267 | 0.0 |
| | RECH | 0.184 | 0.163 | 0.172 | 0.227 | 0.257 | 0.3 |
| | RealGARCH | 0.153 | 0.138 | 0.151 | 0.204 | 0.225 | 1.2 |
| | LSTM-RGARCH | **0.147** | **0.133** | **0.146** | **0.192** | **0.219** | **3.5** |

*Note:* The last column reports the number of times a model has the lowest MSE among the five realized volatility measures. The bottom panel reports the average across the 31 indices. The bold numbers indicate the best scores.

**Table 4**
Forecast performance: MAE for different realized volatility measures.

| | | RV5 | BV | MedRV | RK-Parzen | RSV | Count |
|---|---|---|---|---|---|---|---|
| AEX | GARCH | 0.259 | 0.256 | 0.266 | 0.332 | 0.298 | 0.0 |
| | RECH | 0.237 | 0.233 | 0.242 | 0.300 | 0.276 | 0.0 |
| | RealGARCH | 0.218 | 0.211 | 0.219 | 0.294 | 0.260 | 0.0 |
| | LSTM-RGARCH | **0.209** | **0.202** | **0.209** | **0.274** | **0.249** | **5.0** |
| DJI | GARCH | 0.291 | 0.273 | 0.301 | 0.308 | 0.344 | 0.0 |
| | RECH | 0.267 | 0.253 | 0.278 | 0.281 | 0.323 | 0.0 |
| | RealGARCH | 0.218 | 0.214 | 0.239 | 0.237 | 0.279 | 0.0 |
| | LSTM-RGARCH | **0.208** | **0.204** | **0.226** | **0.223** | **0.272** | **5.0** |
| GDAXI | GARCH | 0.317 | 0.310 | 0.320 | 0.343 | 0.356 | 0.0 |
| | RECH | 0.297 | 0.290 | 0.297 | 0.324 | 0.337 | 0.0 |
| | RealGARCH | 0.237 | 0.233 | 0.242 | 0.274 | 0.284 | 0.0 |
| | LSTM-RGARCH | **0.233** | **0.229** | **0.237** | **0.266** | **0.277** | **5.0** |
| SPX | GARCH | 0.295 | 0.286 | 0.312 | 0.305 | 0.340 | 0.0 |
| | RECH | 0.265 | 0.262 | 0.285 | 0.275 | 0.315 | 0.0 |
| | RealGARCH | 0.222 | 0.225 | 0.247 | 0.240 | 0.277 | 0.0 |
| | LSTM-RGARCH | **0.211** | **0.217** | **0.237** | **0.226** | **0.270** | **5.0** |
| Mean | GARCH | 0.289 | 0.278 | 0.293 | 0.338 | 0.337 | 0.0 |
| | RECH | 0.278 | 0.267 | 0.282 | 0.323 | 0.327 | 0.2 |
| | RealGARCH | 0.248 | 0.238 | 0.258 | 0.305 | 0.302 | 0.7 |
| | LSTM-RGARCH | **0.241** | **0.233** | **0.254** | **0.292** | **0.294** | **4.1** |

*Note:* The last column reports the number of times a model has the lowest MAE among the five realized volatility measures. The last panel reports the average across th 31 indices. The bold numbers indicate the best scores.

2016), i.e., the expected loss is lowest at the true quantile series. The most accurate VaR forecasting model should therefore minimize the quantile loss function.

Although there is no strictly consistent loss function for ES, however, Fissler and Ziegel (2016) found that ES and VaR are jointly elicitable, i.e., there is a class of strictly consistent loss functions for evaluating VaR and ES forecasts jointly. Taylor (2019) show that the negative logarithm of the likelihood function built from the Asymmetric Laplace (AL) distribution is strictly consistent for VaR and ES considered jointly and fits into the class developed by Fissler and Ziegel (2016). This AL based joint loss function is given as

$$\text{JointLoss} := \frac{1}{T_{\text{test}}} \sum_{D_{\text{test}}} \left( -\log\left(\frac{\alpha - 1}{\text{ES}_t^{\alpha}}\right) - \frac{\left(y_t - Q_t^{\alpha}\right)\left(\alpha - I\left(y_t \le Q_t^{\alpha}\right)\right)}{\alpha \text{ES}_t^{\alpha}} \right) \tag{19}$$

with $\text{ES}_t^{\alpha}$ the forecast of $\alpha$-level ES of $y_t$.

To measure the fit of a volatility model to the return series, we also consider the Partial Predictive Score (PPS), which is one of the most commonly used metrics for evaluating predictive performance in statistical modeling. It measures the negative log-likelihood of observing the return series based on our volatility forecast. The model with the smallest PPS is preferred. The PPS score is defined as

$$\text{PPS} := -\frac{1}{T_{\text{test}}} \sum_{D_{\text{test}}} p\left(y_t \mid y_{1:t-1}\right). \tag{20}$$

Table 5 reports the quantile loss, joint loss for 1% and 5% VaR and ES forecast, and the PPS score. The Count panel reports the number of indices where a model achieves the best predictive score. The results show that for most of the indices, the LSTM-RGARCH model produces the best VaR and ES forecasts. LSTM-RGARCH is also superior to its competitors in terms of overall fit to the return series. It reports

**Table 5**
Forecast performance: Tail risk forecast and Partial Predictive Score.

| | | Qloss_1% | JointLoss_1% | Qloss_5% | JointLoss_5% | PPS |
|---|---|---|---|---|---|---|
| AEX | GARCH | 92.549 | 2.462 | 282.507 | 1.909 | 1.342 |
| | RECH | 88.654 | 2.404 | 270.841 | 1.851 | 1.307 |
| | RealGARCH | 86.838 | 2.317 | 273.663 | 1.847 | 1.298 |
| | LSTM-RGARCH | **84.982** | **2.279** | **267.902** | **1.815** | **1.288** |
| DJI | GARCH | **80.691** | 2.340 | 250.738 | 1.767 | 1.175 |
| | RECH | 81.502 | 2.243 | 247.145 | 1.706 | 1.135 |
| | RealGARCH | 82.620 | 2.299 | 242.673 | 1.712 | 1.142 |
| | LSTM-RGARCH | 81.193 | **2.236** | **242.049** | **1.689** | **1.134** |
| GDAXI | GARCH | 100.503 | 2.514 | 316.005 | 2.034 | 1.484 |
| | RECH | **97.312** | **2.480** | **305.339** | **1.991** | **1.453** |
| | RealGARCH | 103.330 | 2.577 | 314.835 | 2.039 | 1.460 |
| | LSTM-RGARCH | 100.372 | 2.557 | 309.042 | 2.016 | 1.453 |
| SPX | GARCH | 83.164 | 2.424 | 253.502 | 1.803 | 1.192 |
| | RECH | 81.716 | 2.327 | 247.041 | 1.735 | 1.149 |
| | RealGARCH | **80.571** | 2.311 | 244.235 | 1.728 | 1.136 |
| | LSTM-RGARCH | 81.334 | **2.287** | **243.364** | **1.714** | **1.135** |
| Mean | GARCH | 78.545 | 2.262 | 260.099 | 1.859 | 1.355 |
| | RECH | 76.549 | 2.224 | 253.029 | 1.825 | 1.339 |
| | RealGARCH | 79.518 | 2.266 | 258.329 | 1.853 | 1.348 |
| | LSTM-RGARCH | **76.141** | **2.223** | **248.735** | **1.812** | **1.336** |
| Count | GARCH | 1 | 3 | 0 | 0 | 0 |
| | RECH | 9 | 7 | 6 | 7 | 11 |
| | RealGARCH | 4 | 3 | 3 | 1 | 4 |
| | LSTM-RGARCH | **17** | **18** | **22** | **23** | **16** |

*Note:* Qloss_1%, JointLoss_1%, Qloss_5%, JointLoss_5% are the quantile loss and jointloss at 1% and 5% respectively. The last two panels report the average scores and the number of times a model has the best predictive scores across the 31 indices.

the lowest PPS on most returns series and on average. Additionally, we find that for most of the metrics, the RECH model performs the best on more indices than RealGARCH. In contrast, in the previous section, RealGARCH dominates RECH regarding forecast error when comparing to the ex-post proxies. This suggests that ranking conditional volatility forecasts by ex-post proxies can sometimes lead to undesirable outcomes, and we should also evaluate predictive performance by economic loss. Regarding the implications of the LSTM-RGARCH model in real financial practice, the model will provide financial practitioners with more accurate and efficient risk forecasts as Table 5 demonstrates. Since the risk coverage capital is determined by the tail risk models employed by the financial institutions and should be proportional to the VaR and ES forecasts, the proposed models are able to increase the profitability of these institutions, by increasing the efficiency of regulatory capital allocation, while still providing sufficient protection against violations to risk forecasts.

Finally, volatility is a also key input in option pricing models such as the Black–Scholes. Accurate volatility forecasts are essential for valuing option contracts, assessing their sensitivities to changes in market conditions, and implementing effective options trading and risk management strategies. Therefore, we also conduct a simulated option trading exercise, to examine the effectiveness of the volatility forecasts produced by the proposed model in option pricing. Appendix 6.1 reports the results.

### 4.4. Statistical significance

The previous sections show that LSTM-RGARCH outperforms the competing models in terms of in-sample fit, forecasting error, tail risk forecast and option pricing. This section tests whether these improvements are statistically significant using the Model Confidence Set (MCS) introduced by Hansen et al. (2011). Let $\mathcal{M}$ be a set of competing models. A set of superior models (SSM) is established under the MCS procedure, which consists of a series of equal predictive accuracy tests given a specific confidence level. Let $L_{i,t}$ be a performance loss, such as the MSE or quantile loss, incurred by model $i \in \mathcal{M}$ at time $t$. Define $d_{i,j,t} = L_{i,t} - L_{j,t}$ to be the relative loss of model $i$ compared to model $j$ at time $t$. The MCS test assumes that $d_{i,j,t}$ is a stationary time series for all $i, j$ in $\mathcal{M}$, i.e., $\mu_{i,j} = \mathbb{E}(d_{i,j,t})$ for all $t$. By testing the equality of the expected loss difference $\mu_{i,j}$, MCS determines if all models have the same level of predictive accuracy. The null hypothesis is

$$H_0 : \mu_{i,j} = 0, \quad \text{for all } i, j \in \mathcal{M}. \tag{21}$$

A model is eliminated when the null hypothesis $H_0$ of equal forecasting ability is rejected. The collection of models that do not reject the null hypothesis $H_0$ is then defined as the SSM. For each model $i \in \mathcal{M}$, the MCS produces a $p$-value $p_i$. The lower the $p$-value of a model, the less likely that it will be included in the SSM. See Hansen et al. (2011) for more details.

Table 6 reports the model confidence sets computed by all the predictive scores. For each model, we report the total number of times, across the 31 indices, that the model is included in the MCS and its average $p$-value (in parentheses). We note that a low $p$-value indicates that the model is unlikely to be the best model and a lower confidence level imposes a stricter selection criterion for the SSM (Hansen et al., 2011). Thus we adopt a strict confidence level (75%) to ensure that only models demonstrating genuinely competitive performance are included in the SSM. At a this confidence level and across the predictive scores, the LSTM-RGARCH models are included in SSM for 24 indices, and have the highest $p$-value of 1 for 20 indices. The result shows that LSTM-RGARCH is the most likely to be included in the SSM and have statistically significant superiority over the other models in all the considered predictive metrics: forecasting error (MSE and MAE), fit to return series (PPS), risk forecast (quantile loss and joint loss) and option pricing.

## 5. Conclusions

The paper presents a novel methodology for volatility modeling, effectively harnessing the strengths of NN and the invaluable information from realized volatility data. Bayesian inference and forecasting are performed using Sequential Monte Carlo with likelihood and data annealing. The LSTM-RGARCH model is evaluated on 31 major world stock markets, demonstrating improved performance in statistical criteria (in-sample fit, forecast error to realized measures) and economic

**Table 6**
Statistical signficance: The number of times a model is included in the set of superior models and the average $p$-value (in parentheses), across the 31 indices.

| | GARCH | RECH | RealGARCH | LSTM-RGARCH |
|---|---|---|---|---|
| MSE | 1 (0.01) | 5 (0.10) | 18 (0.48) | 24 (0.67) |
| MAE | 1 (0.02) | 4 (0.09) | 11 (0.27) | 26 (0.84) |
| Qloss_1% | 9 (0.16) | 20 (0.51) | 12 (0.25) | 22 (0.65) |
| Qloss_5% | 1 (0.01) | 12 (0.30) | 6 (0.15) | 25 (0.80) |
| JointLoss_1% | 12 (0.25) | 16 (0.40) | 14 (0.30) | 24 (0.70) |
| JointLoss_5% | 2 (0.03) | 13 (0.31) | 6 (0.12) | 25 (0.80) |
| PPS | 7 (0.13) | 16 (0.43) | 10 (0.24) | 22 (0.63) |
| OptTrading | 5 (0.10) | 13 (0.39) | 6 (0.14) | 22 (0.70) |

criteria (tail risk forecast and option pricing) compared to the standard GARCH, RECH, and RealGARCH models.

Since financial markets often exhibit nonlinear relationships and complex dependencies that traditional econometric models may struggle to capture, the incorporation of flexible modeling techniques and high-frequency realized measures in LSTM-RGARCH generates a novel framework for capturing the complex dynamics of financial market uncertainty. Our empirical findings indicate the presence of long-term and nonlinear effects of historical market uncertainty on its current and future volatility. This can have significant economic consequences, affecting investor behavior, corporate decisions, consumer spending, and policy responses.

Our proposed framework can be extended in several ways. First, it is possible to include multiple realized volatility measures since Hansen and Huang (2016) discovered that including multiple realized volatility measures evidently improves both the in-sample and out-of-sample fit. The multiple measurement equations could then be replaced by a single LSTM with multiple outputs allowing us to capture the nonlinear relationship between conditional volatility and realized volatility measures as well as the interactions between the realized volatility measures. Second, incorporating financial news into the input of LSTM-RGARCH is an interesting topic to study, as news has been shown to have a major influence on volatility movement (Atkins et al., 2018; Xing et al., 2019; Rahimikia et al., 2021). Last, incorporating transfer learning Thrun and Pratt (1998), Pan and Yang (2010-10) would be a natural extension. Transfer learning can help mitigate the problem of insufficient training data and enable us to use a large number of pre-trained deep learning models and traditional financial econometric models for volatility modeling.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.econmod.2024.106922.

## Data availability

Details on the data are included in the paper.

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